Emergency Department Problem

Scenario

A hospital is reviewing staffing levels in its Emergency Department (ED).

Recently:

- Waiting times have increased
- Staff feel under pressure
- Managers are concerned about costs

Patients arrive throughout the day needing treatment. Some cases are quick to resolve, while others require more time and attention.

The department is staffed by a team of doctors and nurses. Staff work together to assess and treat patients.

Hospital management want to understand:

- What changes might improve performance while controlling cost.

They are open to analysis but are not yet sure what kind of modelling or data is needed.

You have been asked to investigate.

Part A — Structuring the Problem (25 min)

Before building a model, we need to understand the situation clearly.

We do this by identifying:

1. What choice someone needs to make

2. What questions must be answered to support that choice

What do we mean by a “decision”?

A **decision** is a choice between options.

Example (not related to hospitals):

A university is deciding whether to extend library opening hours.

Decision:

- Should we extend library hours, and by how much?

To help make that decision we might need to know:

- How many students use the library late?
- What it costs to keep it open?
- Whether alternative study spaces exist?

Those are **questions we analyse** to support the decision.

A1. What kinds of decisions or changes might management be considering?

A decision is something someone can choose to change.

Look back at the hospital description.

What choices can management make?

Write out as questions.

_____?

_____?

_____?

_____?

There is no single correct wording.

A2. What questions would we need to answer to help make that decision?

These are **analytical questions**.

They are not decisions themselves, they are questions that require data, measurement, or modelling.

These questions help us decide what information or modelling we need before making the decision.

List at least 4.

1.

2.

3.

4.

5.

6.

A3. What analysis would you carry out throw light on the problem?

A4. What data is required and how will you obtain it?

A5. What features of the system might affect what happens?

List anything that might influence how the department performs.

1.

2.

3.

4.

5.

6.

A6. Sketch how you think the system works

Draw how patients might move through the department.

You may include:

- queues
- stages
- staff
- decision points

You may want to do this on a piece of paper. I have left room here for you to put it in the document if you wish.

Part B — Before We Model: Make Predictions (10 min)

We will later explore how staffing affects the system.

Before seeing any model, write down your expectations.

B1. If staffing levels increase...

What do you expect to happen to waiting times?

☐ decrease

☐ stay the same

☐ decrease then level off

☐ unsure

Explain why:

B2. Could delays still happen even if we add more doctors?

List possible reasons, not limited to 2.

1.

2.

B3. What kinds of changes might improve the system?

You can suggest anything, not limited to 2:

1.

2.

Part C — As the Model Is Built

As we build and run the simulation, record your observations.

C1. First simple model

All patients are treated in the same way.

What seems unrealistic?

C2. Adding more detail

What additional features might make the model closer to reality?

1.

2.

3.

4.

C3. When cases differ in complexity or importance...

How might that affect queues or waiting?

C4. If some cases should be handled sooner than others...

How might the system decide what to do?

Describe one possible rule.

Part D — Results from Simulation

You can access the full simulation that we have just built on the module. Use this to fill out the table and the rest of this section.

D1. What staffing level seems best?

Explain your reasoning.

D2. What trade-offs do you notice?

What improves?

What gets worse?

D3. Where does the system seem to struggle most?

Explain briefly.

Part E — Reflection

E1. What surprised you?

E2. Why might simulation be useful for problems like this?

E3. What does our model ignore?

List at least 3 things.

1.

2.

3.

E4. If this were a real hospital...

What data would you want before making recommendations?